Chained transfer differential reference

2026-11-11 · fixed F_{101} fixture · producer-run development checks

Standing: unverified. The reference arithmetic predates this work and is imported unchanged, but both implementations are run by this producer and use Cantor reduction. Agreement is not signed independent verification.

Pinned source and models

Native source: isogeny-labs/crypto commit aabae1979457c3ba9e5112487a907b50cab15a99, package research/chained_jacobian_transfers_20261110/native. The reference is suite/src/prime_hyperelliptic from cryptanalysis commit 68594c8874c75a0a6c04a3874adb1374088ca4f5. source-pin.json and SHA-256 manifests bind all imported files; check.sh refuses drift before compiling.

The source is $C: y^2 = x^6 + 1$ over F_{101}, based at (10,0). The target is $C_2: w^2 = 41z^6 + 9z^4 + 9z^2 + 41$. The order-17 elliptic subgroup has P=(4,41) on $E: y^2 = x^3 + 1$. A faithful coordinate change makes each odd quintic monic: $t = a \cdot z$, $Y = a^2 \cdot Z$ with a=inverse(leading(F)). The reference therefore computes on the same curves, not an unrecorded twist. [pcc:unverified]

Exact development coverage

Check	Retained scope
Source arithmetic	10,404 canonical additions and inverses
Richelot subgroup	All 289 additions in the order-17 target subgroup
Full forward image	10,404 source images and fixed-translation additions; 2,601 distinct target classes
Point-count order	Both Jacobian orders recomputed as 10,404 from F_p/F_{p^2} counts
Successor	Both9,216-class sets: paired additions, dual norms, independent point-count order9,216 and self-inverse counts4/16
Local suite	28 tests pass; 1 pre-existing benchmark remains ignored

Raw outputs retain the original baseline and successor development/pinned receipts. The latest full replay includes source guards, formatting and strict Clippy against the published aabae197 pin. Initial18db046 metadata is archived in source-pins/. PCC-20261111-820d6f remains unsigned/unverified. First Clippy failure and the subsequent pass are retained; its compatibility annotation affects only the unchanged reference's from_fp(&self) API. No correctness test was suppressed. [pcc:unverified]

Reproduce and limits

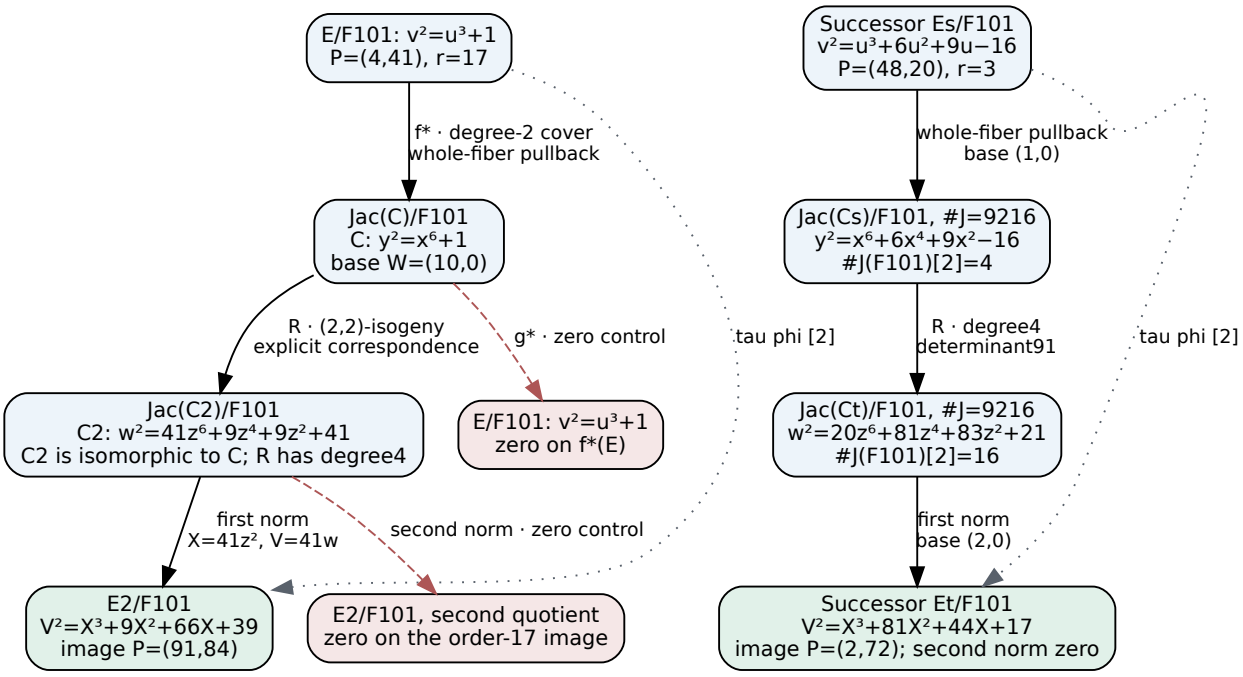
Check out the two repositories as siblings, with the pinned native files in crypto, then run experiments/chained-transfer-reference/check.sh from cryptanalysis. Cargo.lock and numeric dependency versions are fixed.

There is no DLP timing or speed claim, new curve-family coverage, or full formal Picard construction in this record. The initial target is isomorphic to its source. The successor rational 2-torsion counts4/16 obstruct an F101-isomorphism of the Jacobians; its source elliptic quotient has96 points and preserved prime subgroup3. The upstream report retains vanishing projections and direct elliptic shortcuts for both complete endpoint functions. These checks do not establish a new computational reduction. [pcc:unverified]

Primary sources: Galbraith, corrected book ch.8 §8.3 and ch.10 §§10.3–10.4; Kunzweiler, DCC92 (2024), DOI10.1007/s10623-024-01366-1, §§3.1–3.2. The complete source receipt is pinned upstream. KB and vault gaps remain explicit; no novelty claim is made.

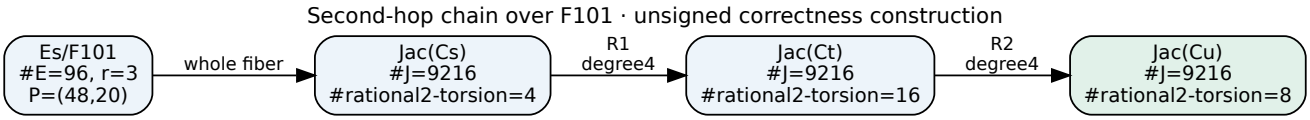
Route of the pinned upstream fixture

F101 chained Jacobian fixture · producer development evidence · PCC unverified



This diagram is copied from the pinned upstream commit. Solid arrows show the explicit maps; dashed arrows are zero controls; dotted arrows are complete finite elliptic shortcuts. Both fixed-field fixtures are shown, with their different subgroup sizes visible. All evidence remains unsigned.

Second nonbacktracking Jacobian hop



The predeclared next pairing(2,18),(-2,28),(-18,-28), with leading scale20, has determinant67 and new base11. The third sextic coefficients are [38,9,29,62,86,17,8]. The unchanged BigUint reference checks all9,216 target classes: paired additions, dual norm and self-inverse count8. Fp/Fp² point counts independently give Jacobian order9,216. Thus all three Jacobians are distinguished over F101 by rational2-torsion4,16,8. [pcc:unverified]

The rational composite kernel has8 elements,4 self-inverse; do not confuse this with the geometric degree16. The upstream native test also checks a trivial next-kernel/previous-dual intersection, both dual norms and nonzero r3 transport through both steps. Source P=(48,20) reaches the canonical class u(T)=T²+86T+27,v(T)=34T+7. Reversed chain and source norm give[8], invertible modulo3. None of these point computations reveals its unknown DLP scalar. [pcc:unverified]

The source guards, formatting, strict Clippy and28-test/1ignored replay passed against aabae197. Receipts second-hop-development-01.txt and second-hop-pinned-01.txt are retained; PCC-20261111-d0f154 is unsigned. This checkpoint predates the later small-prime feature, which requires a separate source-pin update. The initial18db046 and successorb280efc baseline metadata remain archived. [pcc:unverified]

There is no timing, measured ECDLP improvement, cryptographic-size field coverage or full formal Jacobian geometry claim. The upstream queue names checkpoint/budget access gates and proof/merge blockers.